DATA ANALYTICS • STUDENT PROJECT

Supermart Grocery Sales Dashboard

Sales, Profit, Product, Discount and Geographic Performance Analysis

Rs 1.50 Cr	Rs 37.47 L	25.05%	9,994
Total Sales	Total Profit	Profit Margin	Orders

Prepared by: Divyangna

Project platform: SapphireIQ

Dataset: Supermart Grocery Sales, Retail Analytics Dataset (2015 to 2018)

Tools used: Python (pandas, matplotlib) for analysis, Power BI or Excel for the interactive dashboard

1. Project Context and Approach

Ram is working on a Supermart Grocery Sales Dashboard using the company sales data. The management team wants a simple dashboard that helps them understand sales, profit, products, discounts and geographic performance in one place. The goal of this project is to analyse the sales data and convert the findings into an easy to understand dashboard that answers important business questions without going through the complete dataset.

The analysis is organised around four business objectives given in the project brief:

- **Objective 1:** Understand overall sales performance
- **Objective 2:** Understand product performance
- **Objective 3:** Understand profit and discount
- **Objective 4:** Understand geographic performance

2. Dataset Overview

Item	Detail
Records	9,994 order lines
Period covered	02 Jan 2015 to 30 Dec 2018
Columns	Order ID, Customer Name, Category, Sub Category, City, Order Date, Region, Sales, Discount, Profit, State
Categories / Sub Categories	7 / 23
Cities / Regions / States	24 / 5 / 1 (Tamil Nadu)
Customers	50 unique customer names
Currency	Indian Rupees (assumed, as the data is from Tamil Nadu)

3. Data Cleaning and Checks

- **Missing values:** none found in any of the 11 columns.
- **Duplicates:** no duplicate rows found.
- **Order Date:** the column mixed two formats (for example 11-08-2017 and 4/15/2018). Both were converted into one proper date format. Values with dashes were read as day, month, year and values with slashes as month, day, year. Yearly totals are not affected by this choice.
- **Derived fields:** Year, Month and Profit Margin % (Profit divided by Sales) were added.
- **Things to know:** the State column has only one value (Tamil Nadu), so state level comparison is not possible and city level analysis is used instead. The North region has just one order (Rs 1,254), so it is treated as a data limitation and not as a real market.
- **Products:** the data has no individual product name, so Sub Category is used as the product level in Objective 2.

4. Dashboard Snapshot

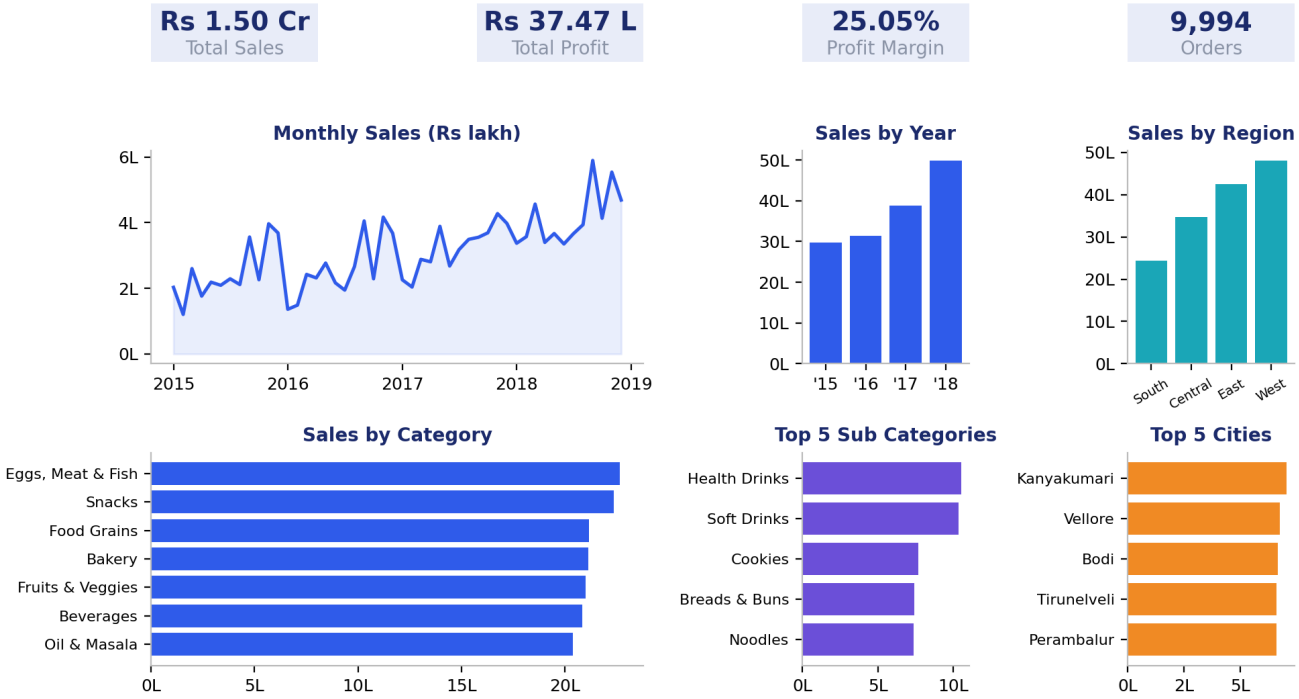


Figure 1: Summary view of the dashboard covering KPIs, trends, categories, products and cities.

OBJECTIVE 1

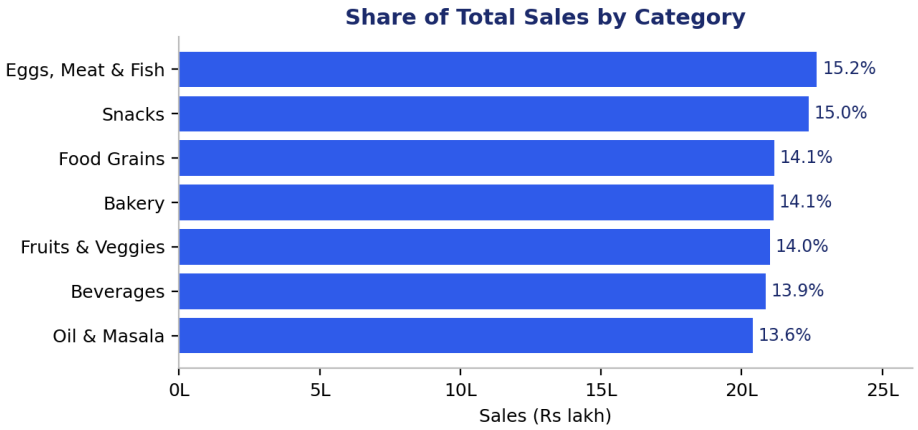
Understand Overall Sales Performance

Business question: How is the overall grocery business performing?

Task 1: Total sales and total profit

KPI	Value
Total Sales	Rs 1,49,56,982
Total Profit	Rs 37,47,121
Profit Margin	25.05%
Total Orders	9,994
Average Sales per Order	Rs 1,497
Average Profit per Order	Rs 375
Average Discount	22.7%

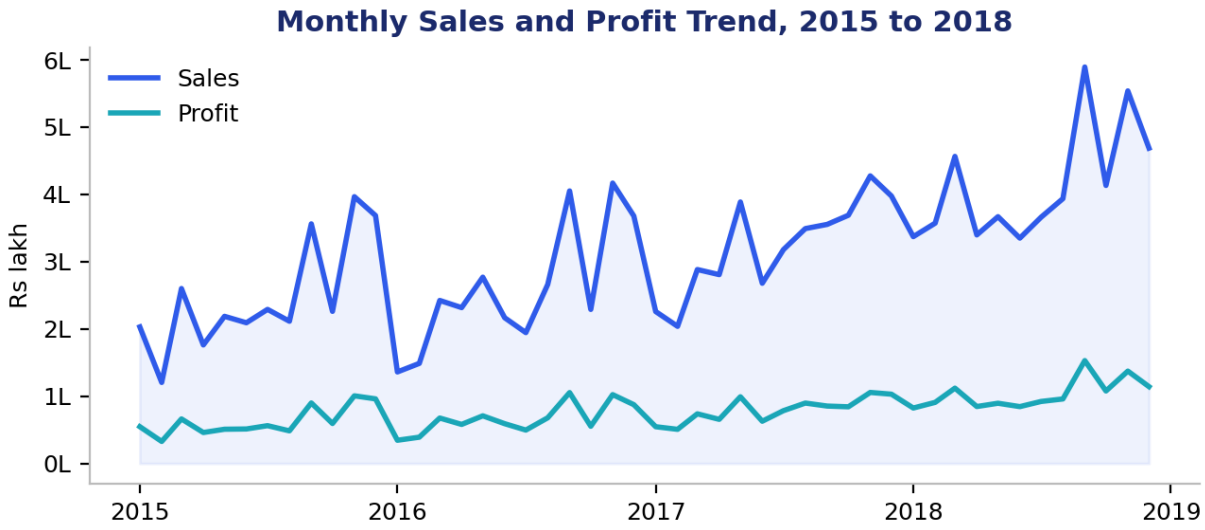
Task 2: Product categories and share of total sales

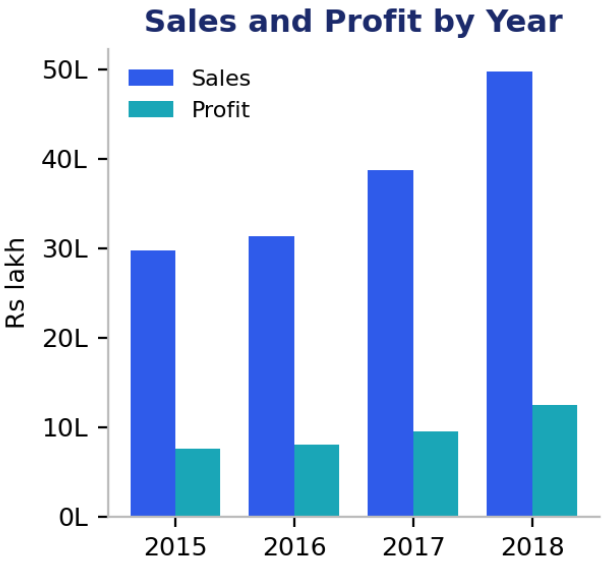


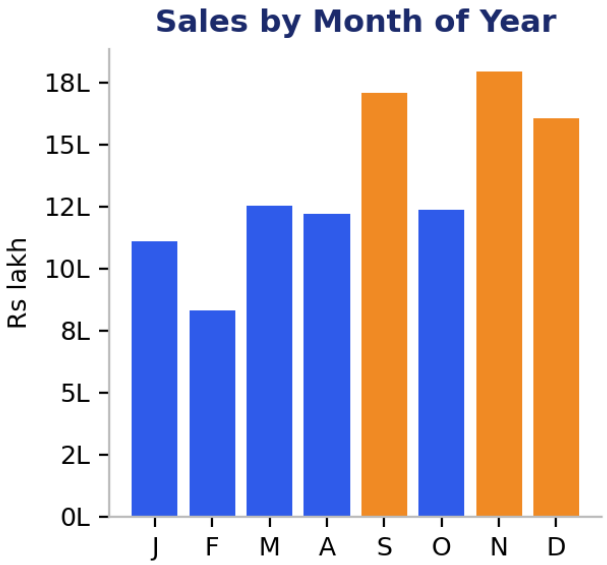
Insight

- **Eggs, Meat & Fish** contributes the largest share at 15.2% (Rs 22,67,401), followed by **Snacks** at 15.0%.
- The spread is narrow. The smallest category, Oil & Masala, still holds 13.6%, so the business is well balanced and not dependent on one category.

Task 3: How sales change over time





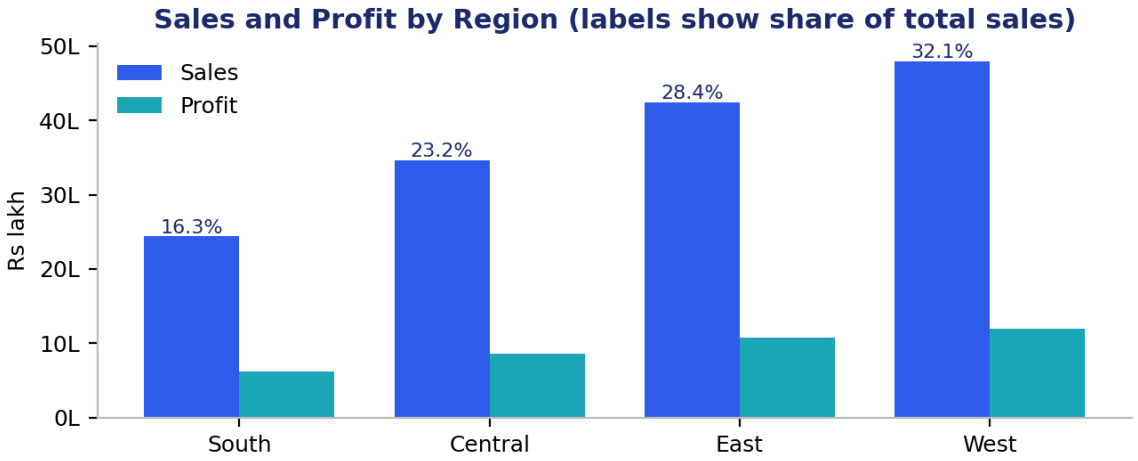


Year	Sales	Profit	Margin	Sales growth
2015	Rs 29,75,599	Rs 7,52,529	25.3%	
2016	Rs 31,31,959	Rs 7,97,193	25.5%	+5.3%
2017	Rs 38,71,912	Rs 9,53,216	24.6%	+23.6%
2018	Rs 49,77,512	Rs 12,44,183	25.0%	+28.6%

Insight

- Sales grew every year, from Rs 29,75,599 in 2015 to Rs 49,77,512 in 2018 (up 67% overall). 2018 grew 28.6% over 2017.
- Profit margin stayed steady near 25% each year, so growth came from higher volume and not from higher margin.
- Seasonality is clear: **November, September, December** are the strongest months, while February and January are the weakest.

Task 4: Regions with the highest sales



Region	Sales	Share	Profit	Margin	Orders
West	Rs 47,98,743	32.1%	Rs 11,92,005	24.8%	3,203
East	Rs 42,48,368	28.4%	Rs 10,74,346	25.3%	2,848
Central	Rs 34,68,156	23.2%	Rs 8,56,807	24.7%	2,323
South	Rs 24,40,461	16.3%	Rs 6,23,563	25.6%	1,619
North	Rs 1,254	0.0%	Rs 401	32.0%	1

Insight

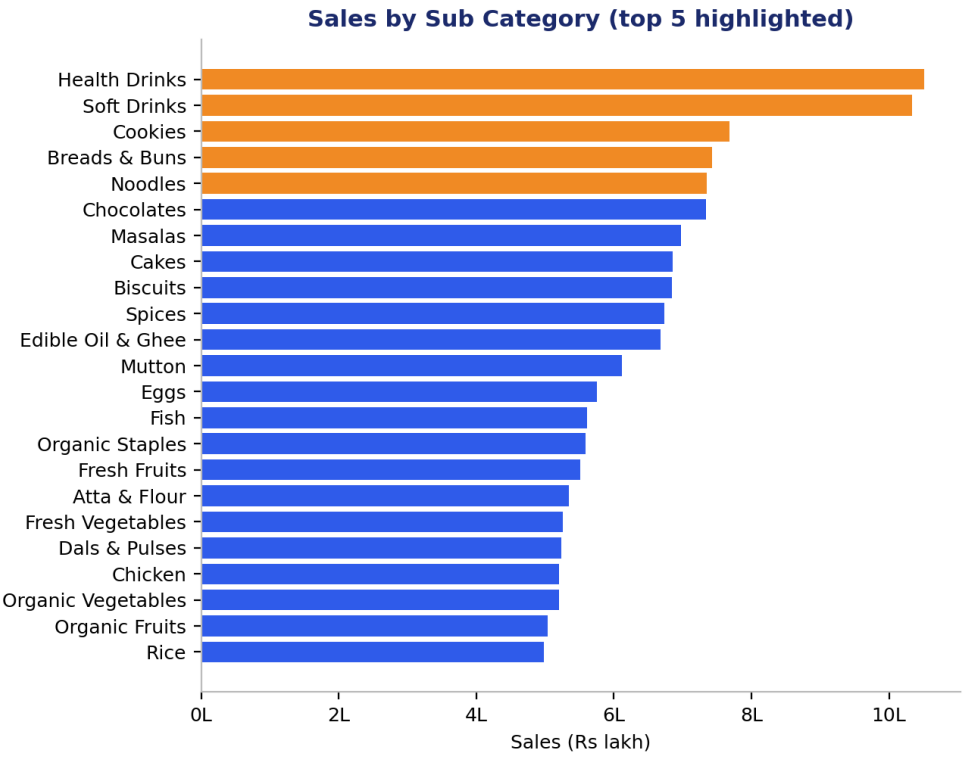
- **West** is the top region with 32.1% of sales, then **East** (28.4%) and **Central** (23.2%). **South** is the smallest real market at 16.3%.
- West and East together bring about 60% of total sales.
- North has only one order, so no conclusion can be drawn for it.

OBJECTIVE 2

Understand Product Performance

Business question: Which products and categories are driving sales and profit?

Task 1: Categories and Sub Categories with the highest sales



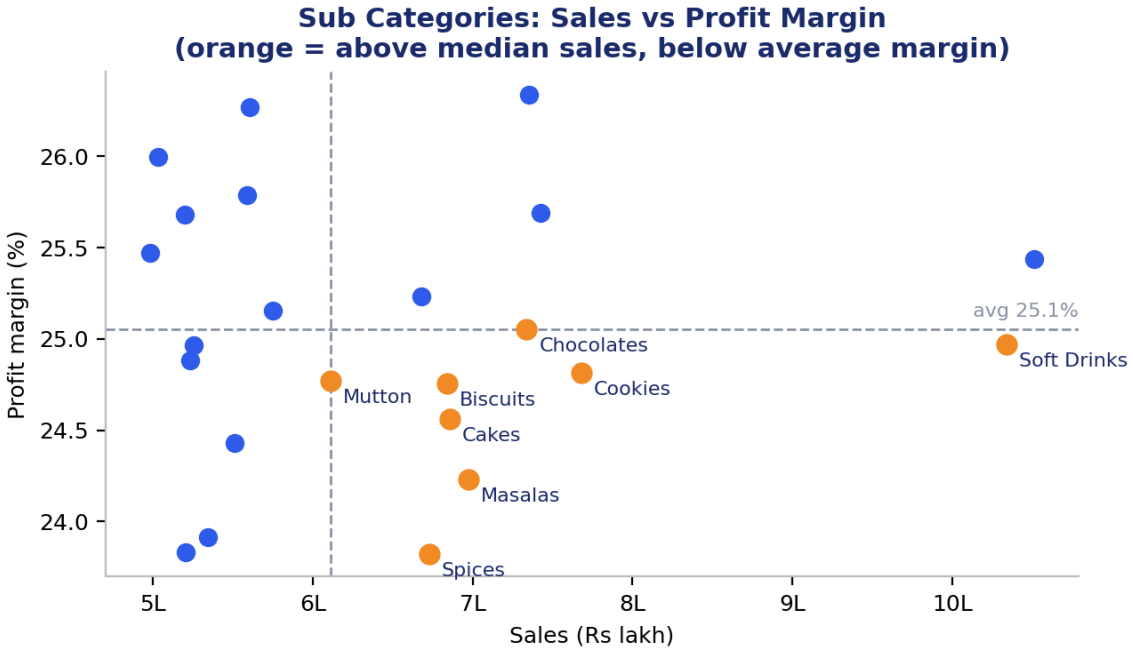
Sub Category (top 8)	Category	Sales	Share	Profit	Margin
Health Drinks	Beverages	Rs 10,51,439	7.0%	Rs 2,67,470	25.4%
Soft Drinks	Beverages	Rs 10,33,874	6.9%	Rs 2,58,136	25.0%
Cookies	Snacks	Rs 7,68,213	5.1%	Rs 1,90,644	24.8%
Breads & Buns	Bakery	Rs 7,42,586	5.0%	Rs 1,90,765	25.7%
Noodles	Snacks	Rs 7,35,435	4.9%	Rs 1,93,686	26.3%
Chocolates	Snacks	Rs 7,33,898	4.9%	Rs 1,83,849	25.1%
Masalas	Oil & Masala	Rs 6,97,480	4.7%	Rs 1,68,999	24.2%
Cakes	Bakery	Rs 6,85,612	4.6%	Rs 1,68,398	24.6%

Insight

- The two biggest Sub Categories are **Health Drinks** (Rs 10,51,439, 7.0%) and **Soft Drinks** (Rs 10,33,874, 6.9%). Both belong to Beverages.
- Beverages is only sixth among categories overall because it has just two Sub Categories, but each of them is much stronger than any other single Sub Category.
- The weakest Sub Categories by sales are Rice, Organic Fruits and Organic Vegetables.

Task 2: Products with high sales but comparatively low profit

Because profit margin is close to 25% almost everywhere, the comparison is made on **profit margin**. A Sub Category is flagged when its sales are above the median (Rs 6,11,200) and its margin is below the overall average of 25.05%.



Flagged Sub Category	Sales	Margin	Gap vs average
Soft Drinks	Rs 10,33,874	24.97%	-0.08 pts
Cookies	Rs 7,68,213	24.82%	-0.24 pts
Chocolates	Rs 7,33,898	25.05%	-0.00 pts
Masalas	Rs 6,97,480	24.23%	-0.82 pts
Cakes	Rs 6,85,612	24.56%	-0.49 pts
Biscuits	Rs 6,84,083	24.76%	-0.30 pts
Spices	Rs 6,72,876	23.82%	-1.23 pts
Mutton	Rs 6,11,200	24.77%	-0.28 pts

Insight

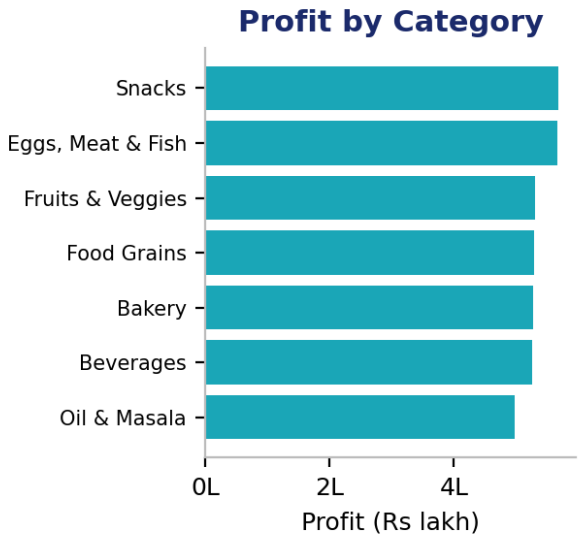
- **8 Sub Categories** have above median sales but below average margin: Soft Drinks, Cookies, Chocolates, Masalas, Cakes, Biscuits, Spices, Mutton.
- **Spices** has the weakest margin among them (23.82%), then Masalas.
- The gaps are small (all within about 1.2 points), so these are watch items for pricing and supplier cost review, not emergencies.

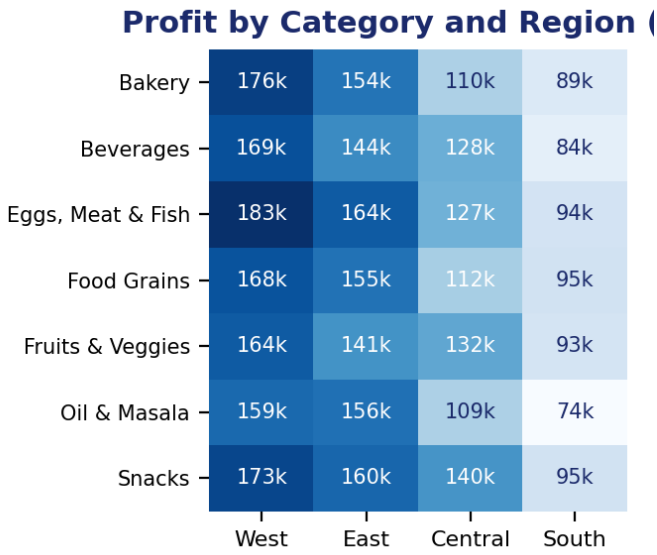
OBJECTIVE 3

Understand Profit and Discount

Business question: How are discounts and profit related to business performance?

Task 1: Categories and regions with the highest profit





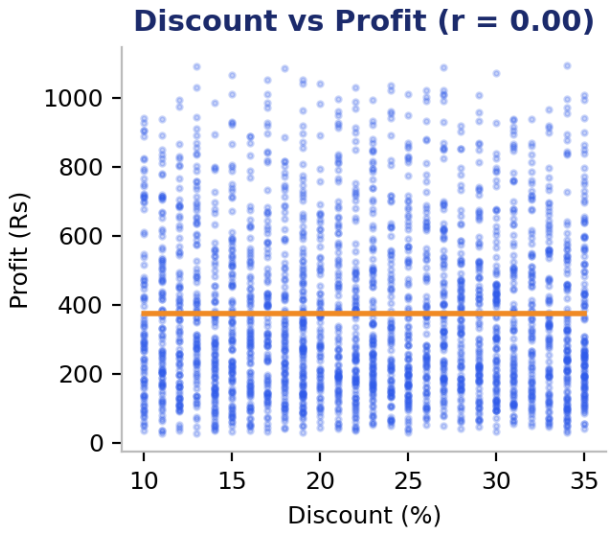
Category	Profit	Margin
Snacks	Rs 5,68,179	25.39%
Eggs, Meat & Fish	Rs 5,67,357	25.02%
Fruits & Veggies	Rs 5,30,400	25.25%
Food Grains	Rs 5,29,163	25.02%
Bakery	Rs 5,28,521	25.02%
Beverages	Rs 5,25,606	25.21%
Oil & Masala	Rs 4,97,895	24.43%

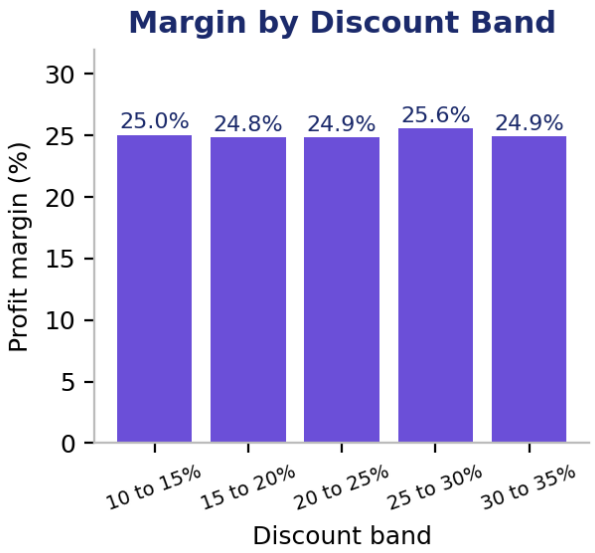
Region	Profit	Share of total profit
West	Rs 11,92,005	31.8%
East	Rs 10,74,346	28.7%
Central	Rs 8,56,807	22.9%
South	Rs 6,23,563	16.6%

Insight

- **Snacks** earns the highest profit (Rs 5,68,179) and also has the best margin (25.39%), narrowly ahead of Eggs, Meat & Fish. **Oil & Masala** has the lowest profit and margin (24.43%).
- **West** generates the most profit (Rs 11,92,005, 31.8% of total). West leads in every category.
- South margin is slightly higher than other regions, but its volume is lowest.

Task 2: Relationship between discount and profit





Discount band	Orders	Avg sales per order	Avg profit per order	Margin
10 to 15%	2,207	Rs 1,496	Rs 375	25.05%
15 to 20%	1,875	Rs 1,502	Rs 373	24.84%
20 to 25%	2,012	Rs 1,502	Rs 373	24.87%
25 to 30%	1,944	Rs 1,485	Rs 380	25.60%
30 to 35%	1,956	Rs 1,499	Rs 373	24.92%

Insight

- There is **no meaningful relationship** between discount and profit. The correlation is 0.00, and the trend line is flat.
- Profit margin is about the same in every discount band (24.8% to 25.6%), so higher discounts are not reducing profit in this data.
- Discounts also do not lift order value: correlation between discount and sales is -0.01, and average sales per order is about Rs 1,500 in every band.
- Profit follows sales instead (correlation 0.61). Growing order value matters more than changing discount levels.
- Caution: the dataset has no cost field, so a discount may still affect the profit calculation outside this data. Test before changing policy.

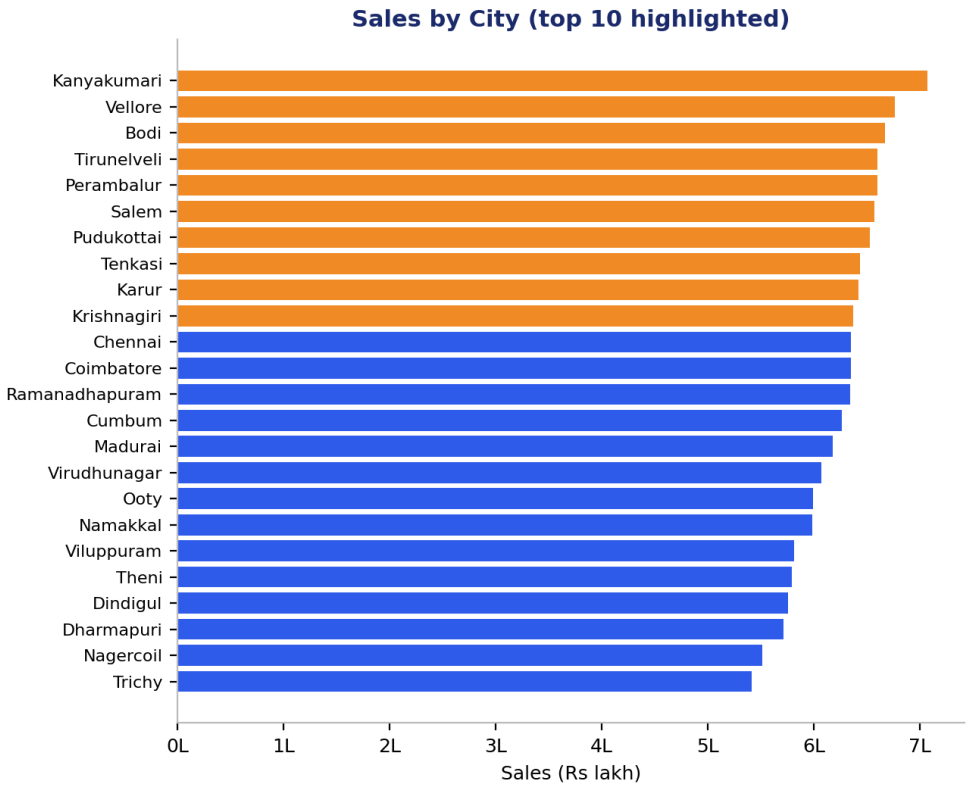
OBJECTIVE 4

Understand Geographic Performance

Business question: Which locations are contributing most to the business?

Task 1: States and cities with the highest sales

All 9,994 orders come from a single state, **Tamil Nadu** (100% of sales), so the comparison is done across its 24 cities.



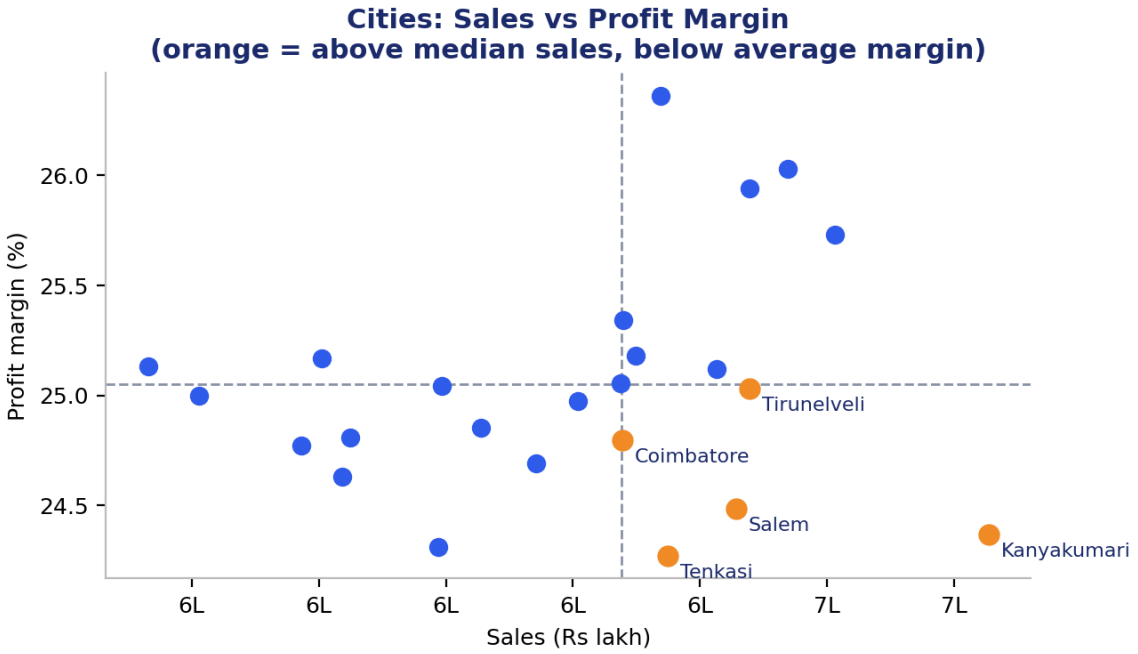
Top 5 cities	Sales	Share	Profit	Margin
Kanyakumari	Rs 7,06,764	4.7%	Rs 1,72,218	24.4%
Vellore	Rs 6,76,550	4.5%	Rs 1,74,073	25.7%
Bodi	Rs 6,67,177	4.5%	Rs 1,73,655	26.0%
Tirunelveli	Rs 6,59,812	4.4%	Rs 1,65,169	25.0%
Perambalur	Rs 6,59,738	4.4%	Rs 1,71,132	25.9%

Insight

- **Kanyakumari** leads with Rs 7,06,764 (4.7%), followed by Vellore, Bodi, Tirunelveli and Perambalur.
- The top 10 cities together hold about 44% of sales. The gap between the best city (Rs 7,06,764) and the lowest (Trichy, Rs 5,41,403) is under 25%, so sales are spread widely.

Task 2: Locations with high sales but comparatively low profit

Cities with sales above the median (Rs 6,34,567) and margin below the overall average (25.05%) are flagged.



Flagged city	Sales	Margin	Gap vs average
Kanyakumari	Rs 7,06,764	24.37%	-0.69 pts
Tirunelveli	Rs 6,59,812	25.03%	-0.02 pts
Salem	Rs 6,57,093	24.49%	-0.57 pts
Tenkasi	Rs 6,43,652	24.27%	-0.78 pts
Coimbatore	Rs 6,34,748	24.80%	-0.26 pts

Insight

- **Kanyakumari, Tirunelveli, Salem, Tenkasi, Coimbatore** have high sales but below average margin. **Kanyakumari** is the biggest city by sales and also has one of the lowest margins (24.37%).
- Cities such as Vellore, Bodi and Karur combine high sales with margins above 25.7%, and can serve as benchmarks for pricing and cost practice.

5. Key Findings

- The business earned **Rs 1,49,56,982** in sales and **Rs 37,47,121** in profit, a steady margin of 25.1%.
- Sales grew every year and were up 67% from 2015 to 2018. Peak months are November, September, December.
- All 7 categories contribute 13.6% to 15.2% each, a balanced mix. Eggs, Meat & Fish leads in sales and Snacks leads in profit.
- Health Drinks and Soft Drinks are the two strongest Sub Categories.
- West and East bring about 60% of sales. South is the smallest region.
- Discount level shows no link with profit or order value.
- Sales are spread evenly across 24 Tamil Nadu cities, with Kanyakumari on top.

6. Recommendations

- **Plan stock and promotions around peak months** (September, November, December) and run campaigns to lift the slow months of January and February.
- **Protect and grow Beverages:** keep Health Drinks and Soft Drinks well stocked, as they are the highest selling Sub Categories.
- **Review margins of flagged items** (Soft Drinks, Cookies, Masalas, Spices, Cakes, Biscuits, Mutton, Chocolates) through supplier cost or small price changes.
- **Review discount strategy:** since discounts of 30 to 35% do not give higher profit or bigger orders than 10 to 15%, test smaller discounts on selected products and track the result.
- **Expand in South** where sales are lowest but margin is the highest among regions, and learn from West, which leads in every category.
- **Review flagged cities** (Kanyakumari, Tirunelveli, Salem, Tenkasi, Coimbatore) for local costs and pricing.
- **Fix data quality:** use one date format, add a real State value for each city and product names, and include cost data so discount impact can be measured fully.

7. How to Build the Interactive Dashboard (Power BI or Excel)

A PDF cannot be interactive, so the findings above are shown as charts. To produce the live dashboard that management asked for, follow these steps.

Step	Action
1	Load the CSV. In Power Query, convert Order Date into a proper date (handle the two formats), set Sales and Profit as numbers, and add Year, Month and Profit Margin columns.
2	Create measures: Total Sales = SUM(Sales); Total Profit = SUM(Profit); Profit Margin % = DIVIDE([Total Profit],[Total Sales]); Avg Discount = AVERAGE(Discount); Orders = COUNTROWS(table).
3	Top row: KPI cards for Total Sales, Total Profit, Profit Margin %, Orders and Avg Discount.
4	Add visuals: line chart of sales by month, column chart by Year, bar chart by Category and Sub Category, column chart by Region, bar chart by City, and a scatter chart of Discount against Profit.
5	Add slicers for Year, Region, Category and City so management can filter every chart at once.
6	Use one colour theme, clear titles, and arrange the page in this order: KPIs, trend, products, discount, geography. Add one page per objective if needed.
7	Take screenshots of each page and add them to this report, then upload.

Conclusion. Supermart Grocery is a growing, healthy business with a stable 25% margin, balanced categories and broad geographic spread. Growth opportunities lie in seasonal planning, strengthening beverage and snack

sales, improving margins on the flagged products and cities, and rethinking blanket discounts.